LabGenius S Series Transfer Robot

Vision Module Manual



- Version：V1.0
- Applicable Models：LabGenius S Series Transfer Robots

Bioyond Robotics(Shanghai)Co., Ltd

Contents

1. PREFACE 2

1.1. COPYRIGHT STATEMENT 2
1.2. SCOPE OF APPLICATION 2

2. INSTALLATION 3

2.1. SOFTWARE INSTALLATION 3

3. CONFIGURATION 4

3.1. LAN CABLE CONFIGURATION 4
3.2. SYSTEM CONFIGURATION 4

4. CALIBRATION 5

4.1. CALIBRATION BOARD PLACEMENT POSITION 5
4.2. VISION MODULE DEMO 6

1. Preface

Dear User,

Thank you for choosing the LabGenius S Series Transfer Robot manufactured by Bioyond Robotics(Shanghai) Co., Ltd.

This manual aims to help you quickly master the usage of the vision module, enabling efficient interaction with the robot. Please read the relevant materials carefully to make the most out of this product.

If you have any questions or needs during use, please feel free to contact us via phone or email, or visit our website. We are committed to providing you with the best service.

1.1. Copyright Statement

- All photos and illustrations mentioned in this publication are the property of Bioyond Robotics(Shanghai) Co., Ltd, and are protected by law.
- The information in this document is subject to change without notice. The manufacturer makes no representations or warranties, express or implied, regarding the content of this document for a particular purpose.
- Without prior written permission, this document may not be reproduced, copied, or translated into other languages.
- Contents herein may be revised without further notice.
- In case of discrepancies between illustrations and actual product appearances, the actual product prevails.
- This manual does not represent any specifications or hardware/software configurations of the product. For detailed specifications and configurations, please refer to the relevant product agreements, packing list, or consult with the sales department.
- Every effort has been made to ensure the accuracy and completeness of the contents in this manual. However, errors or omissions may still exist. Bioyond Robotics(Shanghai) Co., Ltd. continuously strives to optimize and improve its products and services. Therefore, we reserve the right to modify the product and its description in this manual at any time without prior notice. If you find any discrepancies between the actual product and the manual or need the latest information, please feel free to contact us or visit our website for inquiries.

1.2. Scope of Application

- This manual applies exclusively to LabGenius S Series Transfer Robots manufactured by Bioyond Robotics(Shanghai) Co., Lt

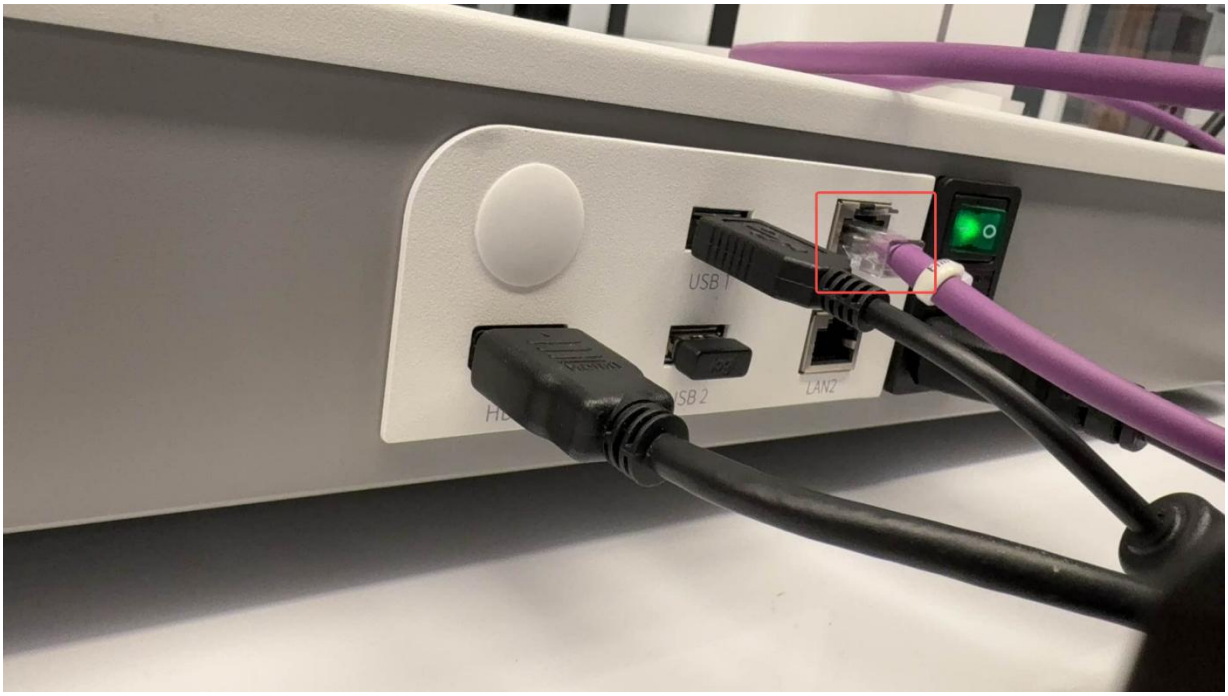
2. Installation

2.1. Software Installation

Before using the vision module, you need to install the MVS driver on the computer running the vision server. Follow the prompts to complete the installation step by step.

3. Configuration

3.1. LAN Cable Configuration



For Polar, ensure the network cable is plugged into the upper LAN port so that the device can communicate with both the camera and the robot simultaneously.

For the S-series robot, you need to use a switch to connect the computer, the network port on the top of the robot, and the network port on the bottom of the robot to ensure smooth network communication.

The specific steps are as follows:

1. Connect the computer's network port to one of the switch ports using an Ethernet cable.
2. Connect the network port on the top of the robot to another port on the switch using an Ethernet cable.
3. Connect the network port on the bottom of the robot to another port on the switch using an Ethernet cable.

Set the computer's IP address to 192.168.1.* (for example, 192.168.1.15), with the subnet mask set to 255.255.255.0.

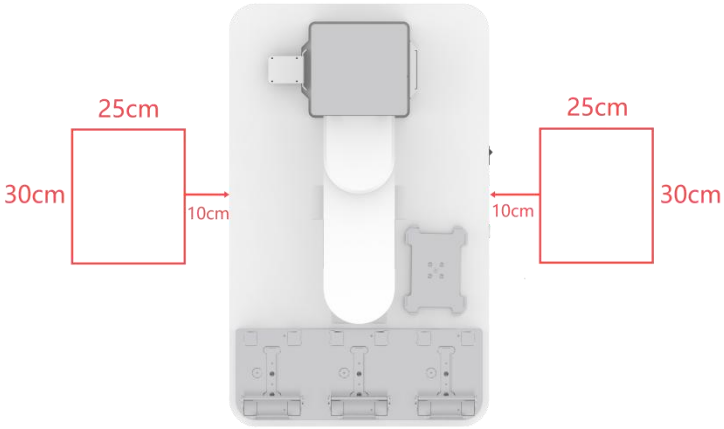
3.2. System Configuration

The region format needs to be set to the United Kingdom.



4. Calibration

4.1. Calibration Board Placement Position



Ensure the calibration board is placed within the red area. The second and third axes should not be too straight. The camera should be positioned about 20-30 cm above the calibration board, with the large circle facing the robot. The following image shows the best practice for the initial position.



4.2. Vision Module Demo

First, run LiveDetection.

```
namespace BR.PC.VisionSDK.Test;

0 个引用
internal class Program
{
    0 个引用
    static async Task Main(string[] args)
    {
        LiveDetection api = new LiveDetection();
        await api.Run();
    }

}
```

Drag the robot and observe the captured photos until the photo shows an appropriate brightness and displays 'Ready', as shown in the image below.



Then run VisionTeach.

```
namespace BR.PC.VisionSDK.Test;

0 个引用
internal class Program
{
    0 个引用
    static async Task Main(string[] args)
    {
        VisionTeach api = new VisionTeach();
        await api.Run();
    }
}
```

The robot will first move up and down, then left and right. After repeating this process, it will proceed to grab the calibration board. The variable finalPose represents the final pose for grabbing the calibration board.

```
if (visionResult.IsVisionFinished)
{
    finalPose = visionResult.Pose;
    break;
}
```

Note: RobStudio and VisionTeach cannot run simultaneously.